

Authentication over Arbitrarily Varying Channels with Causal Adversaries

Abstract

We study authentication over discrete memoryless arbitrarily varying channels (AVCs) when the adversary has access to noiseless, causal feedback of the channel output. We characterize the conditions under which positive authentication rate is achievable for both deterministic and stochastic encoding, and show that the converse is governed by an “overwritability” property of the channel.

I. INTRODUCTION

Authentication is the process by which a receiver, at the end of a communication channel, decodes a transmitted message and declares with confidence that the message has not been tampered with by an adversary; if the receiver cannot make such a declaration, it raises an alarm rather than committing to a possibly compromised decoding. We consider the authentication problem over an arbitrarily varying channel (AVC) $W_{Y|X,S}$, in which the state S is chosen by an adversary that observes the channel output causally (Fig. 1).

For reliable communication over an AVC, it is shown in [1] that the deterministic capacity is positive iff $W_{Y|X,S}$ is not *symmetrizable*. An analogous characterization for authentication is due to Kosut and Kliewer [2], [3]: the authentication capacity against an oblivious adversary is positive iff $W_{Y|X,S}$ is not *overwritable*, where $W_{Y|X,S}$ is overwritable if there is a conditional distribution $Q_{S|X'}$ such that, for every $x, x' \in \mathcal{X}$,

$$\sum_{s \in \mathcal{S}} Q_{S|X'}(s|x') W(y|x, s) = W(y|x', s_0) \quad \forall y \in \mathcal{Y}. \quad (1)$$

Subsequent work extended this characterization to myopic adversaries with non-causal noisy observations of the input, with $W_{Y|X,S}$ being not I-overwritable [4] and not distribution-overwritable [5] sufficient for positive authentication capacity in the corresponding settings.

In this paper, we consider an *output-feedback* adversary that, at each time t , chooses the state S_t as a function of (S^{t-1}, Y^{t-1}) . This is the causal counterpart of the omniscient adversary and refines the oblivious adversary of [2]. The corresponding causal-adversary AVC reliability problem is treated in [6]–[9].

Main results

We characterize when authentication has positive capacity against this adversary for both deterministic and stochastic codes. Our results are stated in terms of two channel conditions strictly weaker than (1), called *distribution-* and *I-overwritability* (formalized in Section III). We show that:

- (i) the stochastic authentication capacity is positive iff $W_{Y|X,S}$ is not distribution-overwritable;
- (ii) the deterministic authentication capacity is positive iff $W_{Y|X,S}$ is not I-overwritable;
- (iii) whenever the stochastic authentication capacity is positive, it equals the no-adversary Shannon capacity C_{s_0} .

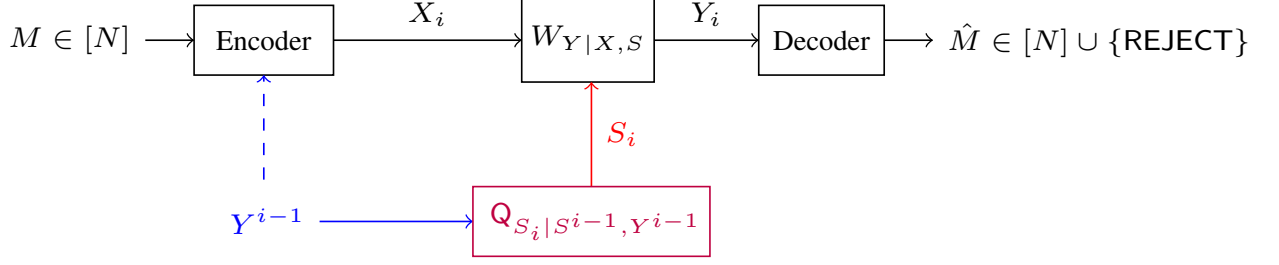


Fig. 1: Channel model. The adversary observes the channel output Y^{i-1} causally and selects state S_i . The dashed feedback link to the encoder is optional.

TABLE I: Authentication-capacity dichotomies across adversary models.

Adversary's view	$C_{\text{auth, stoch}} = 0$ iff	$C_{\text{auth, det}} = 0$ iff	Ref.
None (oblivious)	oblivious-OW	oblivious-OW	[2]
Y^{t-1} (causal f.b.)	distribution-OW	I-OW	Thms. 7, 9
X^n (omniscient)	I-OW	I-OW	[4]

The deterministic-code capacity, when positive, remains open, as it does in the omniscient setting [4]. Table I places these results alongside the oblivious [2] and omniscient dichotomies. Our converse is established through a wait-and-overwrite attack based on a stopping rule constructed from the posterior guessing probability of the message; the achievability uses the authentication-tag construction of [5], modified for feedback adversaries through a martingale concentration argument.

Related work

Authentication has been studied in many settings beyond the present adversary model. The problem originates in cryptography [10]–[12], and has been considered on noisy channels with shared keys, in keyless schemes, and in Byzantine multiple-access settings [13]–[19]. Within the AVC framework, variants of the Kosut–Kliewer characterization include AWGN authentication [20], structured codes [21], and additional myopic and multiple-access models [22], [23]. Two-phase constructions in this line are related to identification codes [24]–[27], which also underpin our achievability through the doubly-exponential message growth of the authentication tag. The wait-and-overwrite converse is connected to the line of adversarial hypothesis testing [28]–[31].

Organization

Section II gives preliminaries; Section III states the formal results; Section IV the converse; Sections V and VI the achievability for stochastic and deterministic codes; Section VII channels that separate the three overwritability classes. Detailed proofs are in this paper.

II. PRELIMINARIES

Notation: For a positive integer N , $[N] \triangleq \{1, \dots, N\}$. Script letters $\mathcal{X}, \mathcal{Y}, \mathcal{S}, \mathcal{M}, \dots$ denote finite alphabets, capital letters the corresponding random variables, and lowercase letters their realizations. We write $y^t = (y_1, \dots, y_t)$. The set of probability distributions on a finite set $\mathcal{A}$ is $\mathcal{P}(\mathcal{A})$; $\mathbf{1}_a$ is the point mass on $a \in \mathcal{A}$ and $\mathbb{V}(\cdot, \cdot)$ is the variational distance.

Channel and adversary: $W_{Y|X,S}$ is a discrete memoryless channel with finite alphabets $\mathcal{X}, \mathcal{S}, \mathcal{Y}$ and a fixed *no-adversary* state $s_0 \in \mathcal{S}$; $W_{Y|X,S=s_0}$ is the legitimate channel and C_{s_0} its Shannon capacity. At time t the encoder emits $X_t \in \mathcal{X}$, the adversary picks $S_t \in \mathcal{S}$, and $Y_t \sim W(\cdot | X_t, S_t)$. The receiver outputs $\hat{M} \in [N] \cup \{\text{REJECT}\}$, where REJECT flags the adversary's presence. Throughout, $M \in [N]$ is uniformly distributed.

Codes: An (N, n) code is a sequential encoder

$$F = \{P_{X_t|M, X^{t-1}, Y^{t-1}}\}_{t=1}^n$$

paired with a decoder $\phi: \mathcal{Y}^n \rightarrow [N] \cup \{\text{REJECT}\}$. The code is *deterministic* if each conditional law is a Dirac mass, and *stochastic* otherwise. The encoder is in the *feedback* (resp. *no-feedback*) class according to whether X_t may depend on Y^{t-1} .

Adversarial strategies and error metric: An *adversarial strategy* is a sequence $Q = \{Q_{S_t|S^{t-1}, Y^{t-1}}\}_t$; we write Q_0 for the trivial strategy $S_t \equiv s_0$. The *authentication error* of a code (F, ϕ) is

$$\varepsilon(F, \phi) \triangleq \Pr_{F, \phi, Q_0}(\hat{M} \neq M) + \sup_Q \Pr_{F, \phi, Q}(\hat{M} \notin \{M, \text{REJECT}\}). \quad (2)$$

A rate R is *achievable* if there exist $(2^{nR}, n)$ codes with $\varepsilon \rightarrow 0$. We write $C_{\text{auth}, \bullet}^\bullet$ for the resulting capacity, where $\bullet \in \{\text{fb}, \text{no-fb}\}$ and $\circ \in \{\text{det}, \text{stoch}\}$. Throughout, we let $\epsilon_n \triangleq \varepsilon(F, \phi)$ denote the auth-error of the code under consideration; since the no-adversary error $\Pr_{Q_0}(\hat{M} \neq M)$ is one of the two non-negative terms in (2), ϵ_n also upper-bounds it, and any bound stated in terms of the no-adversary error a fortiori holds with ϵ_n in its place.

Posterior-guessing probability: Our converse relies on the adversary being able to narrow down the set of messages by way of the *guessing probability*, defined as

$$G_{A|B}(b) \triangleq \sum_{a \in \mathcal{A}} \left(\Pr_{A|B}(a | b) \right)^2.$$

Note that $G_{A|B}(b)$ is the probability that two independent draws from the posterior $\Pr(A | B = b)$ coincide. We will use $G_{M|Y^t, S^t}(y^t, s_0^t)$, evaluated under the no-adversary law.

III. MAIN RESULTS

A. Overwritability conditions

The informal classification described in Section I is made precise as follows; each condition states that, at the indicated level of granularity, the adversary can pick states so that the attacked channel matches the no-adversary channel $W(\cdot | x', s_0)$.

Definition 1 (Obviously overwritable [2]). $W_{Y|X,S}$ is *obviously overwritable* if there is a conditional distribution $Q_{S|X'}$ satisfying (1).

Definition 2 (I-overwritable [4]). $W_{Y|X,S}$ is *I-overwritable* if for every $x, x' \in \mathcal{X}$ there is $Q \in \mathcal{P}(\mathcal{S})$ such that

$$\sum_{s \in \mathcal{S}} Q(s) W(y | x, s) = W(y | x', s_0) \quad \forall y \in \mathcal{Y}.$$

Definition 3 (Distribution-overwritable). $W_{Y|X,S}$ is *distribution-overwritable* if for every $P \in \mathcal{P}(\mathcal{X})$ and every $x' \in \mathcal{X}$ there is $Q \in \mathcal{P}(\mathcal{S})$ such that

$$\sum_{x, s} P(x) Q(s) W(y | x, s) = W(y | x', s_0) \quad \forall y \in \mathcal{Y}.$$

Remark 4 (Symmetric form of distribution-overwritability). $W_{Y|X,S}$ is distribution-overwritable iff for every $P, P' \in \mathcal{P}(\mathcal{X})$ there is $Q \in \mathcal{P}(\mathcal{S})$ with $\mathbb{E}_{X,S} W_{Y|X,S}(y | X, S) = \mathbb{E}_{X',S} W_{Y|X,S}(y | X', S)$ for all y , where $(X, S) \sim P \times Q$ and $X' \sim P'$. The forward implication follows by taking convex combinations over x' of the strategies in Definition 3.

Remark 5 (Nesting). Obviously overwritable $\Rightarrow$ distribution-overwritable $\Rightarrow$ I-overwritable. The reverse implications fail in general (Section VII).

Remark 6 (Non-overwritability quantification). Since the alphabets are finite, the involved sets of distributions are compact. Consequently, when $W_{Y|X,S}$ is not distribution-overwritable there exists $\nu > 0$ with

$$\max_{\substack{P \in \mathcal{P}(\mathcal{X}) \\ x' \in \mathcal{X}}} \min_{Q \in \mathcal{P}(\mathcal{S})} \mathbb{V}(\mathbb{E}_{X,S} W(\cdot | X, S), W(\cdot | x', s_0)) \geq \nu; \quad (3)$$

similarly, when $W_{Y|X,S}$ is not I-overwritable there exist $x_0, x'_0 \in \mathcal{X}$ and $\delta > 0$ with

$$\min_{Q \in \mathcal{P}(\mathcal{S})} \mathbb{V}(\mathbb{E}_S W(\cdot | x_0, S), W(\cdot | x'_0, s_0)) \geq \delta. \quad (4)$$

B. Authentication capacity and positivity conditions

Theorem 7 (Positivity for stochastic codes). *The following are equivalent:*

- (i) $C_{\text{auth, stoch}}^{\text{no-fb}} > 0$;
- (ii) $C_{\text{auth, stoch}}^{\text{fb}} > 0$;
- (iii) $W_{Y|X,S}$ is not distribution-overwritable.

Theorem 8 (Stochastic capacity characterization). *Whenever $C_{\text{auth, stoch}}^\bullet > 0$,*

$$C_{\text{auth, stoch}}^\bullet = C_{s_0} \quad \text{for } \bullet \in \{\text{fb}, \text{no-fb}\}.$$

Theorem 9 (Positivity for deterministic codes). *The following are equivalent:*

- (i) $C_{\text{auth, det}}^{\text{no-fb}} > 0$;
- (ii) $C_{\text{auth, det}}^{\text{fb}} > 0$;
- (iii) $W_{Y|X,S}$ is not I-overwritable.

Remark 10. For deterministic codes our results give only positivity. The achievable rate is not known to equal C_{s_0} : after the wait phase, the adversary can guess the intended codeword and act effectively *omniscient*, but the omniscient-adversary capacity for deterministic codes is itself open. Determining $C_{\text{auth, det}}^\bullet$ is left as an open problem.

IV. THE WAIT-AND-OVERWRITE ATTACK

We give a single attack that proves the converse halves for both Theorems 7 and 9. The attack has two phases: a *wait* phase in which the adversary tracks the no-adversary posterior over M and plays $S_t = s_0$, and an *overwrite* phase in which the adversary, after committing to a sampled guess and decoy from the posterior, forces the channel output to mimic a no-adversary transmission of the decoy. Whether the overwriting succeeds depends on which overwritability condition the channel satisfies; hence the same attack yields the converse of Theorem 7 (under distribution-overwritability) and of Theorem 9 (under I-overwritability).

A. The attack

Fix a code (F, ϕ) and a threshold $\gamma \in (0, 1/(32|\mathcal{Y}|^2))$. At each step t during the wait phase the adversary sets $S_t = s_0$ until the no-adversary posterior $G_{M|Y^{t-1}, S^{t-1}}$ (as defined in Section II) exceeds γ . Let T_γ be the first such index. The adversary then samples a guess $\hat{M}$ and a decoy $\tilde{M}$ independently from the posterior at T_γ and, in the overwrite phase, plays a per-step strategy $Q^{(\tau)}$ chosen so that the marginal law of Y_τ matches a no-adversary transmission of $\tilde{M}$ under the encoder's hypothetical use of message $\hat{M}$. The pseudocode is given in Algorithm 1.

Algorithm 1: Wait-and-overwrite attack.

Input: Encoder $\{P_{X_t|M, X^{t-1}, Y^{t-1}}\}$, threshold γ , observations $y_1, \dots, y_n$.

/* Wait phase */

$t \leftarrow 1$;

while $G_{M|Y^{t-1}, S^{t-1}}(y^{t-1}, s_0^{t-1}) \leq \gamma$ **and** $t \leq n$ **do**

$s_t \leftarrow s_0$; $t \leftarrow t + 1$;

end while

$T_\gamma \leftarrow t - 1$; sample $\hat{m}, \tilde{m} \stackrel{\text{i.i.d.}}{\sim} P_{M|Y^{T_\gamma}, S^{T_\gamma}}^{\text{na}}(\cdot | y^{T_\gamma}, s_0^{T_\gamma})$;

/* Overwrite phase */

for $\tau = T_\gamma + 1, \dots, n$ **do**

 Sample $s_\tau \sim Q^{(\tau)}$ so that the marginal at step τ matches a no-adv transmission of $\tilde{m}$;

end for

B. Stopping-time lemma

The wait phase rests on the following stopping-time control of the posterior. Let M be uniform on $[N]$ and Y^n be the channel outputs under $S^n = s_0^n$.

Lemma 11 (Stopping time). *Fix $\gamma \in (0, 1/(32|\mathcal{Y}|^2))$ and let*

$$T_\gamma \triangleq \inf \{t \in [n] : G_{M|Y^t, S^t}(Y^t, s_0^t) > \gamma\},$$

with the convention $\inf \emptyset = \infty$. For any (N, n) stochastic feedback code with no-adversary error probability p_e ,

$$\Pr(T_\gamma > n \mid S^n = s_0^n) \leq \frac{p_e}{1 - \sqrt{\gamma}}, \quad (5)$$

$$\Pr(G_{M|Y^{T_\gamma}, S^{T_\gamma}} > \frac{1}{2} \mid T_\gamma \leq n, S^{T_\gamma} = s_0^{T_\gamma}) \leq \sqrt{2} |\mathcal{Y}| \sqrt{\gamma} < \frac{1}{4}. \quad (6)$$

In the above, (5) bounds the non-stopping probability by $p_e/(1 - \sqrt{\gamma})$. Note that the reliability of the code ($p_e \rightarrow 0$) guarantees that the receiver, and therefore, the adversary observing the same wait-phase outputs, accumulates evidence about M . (6) asserts that, on the stopping event, the posterior at T_γ exceeds $1/2$ with probability strictly less than $1/4$. This non-degeneracy enables the symmetry driving both converses: the adversary's sampled $\hat{M}$ and decoy $\tilde{M}$ satisfy $\hat{M} \neq M$ and $\tilde{M} \neq M$ with conditional probability $\geq 1/2$, so the attack succeeds with constant probability for both stochastic and deterministic codes.

Proof sketch: For (5), $g(y^n) \geq (1 - p^*(y^n))^2$ where p^* is the conditional MAP error, so $g(Y^n) \leq \gamma$ forces $p^*(Y^n) \geq 1 - \sqrt{\gamma}$, and Markov on $\mathbb{E}[p^*(Y^n)] \leq p_e$ gives the bound. For (6), on $\{T_\gamma = t\}$ minimality gives $g(Y^{t-1}) \leq \gamma$, and a sub-multiplicative bound on $\sqrt{G}$ (Appendix ??) combined with Markov yields the displayed inequality. Full proof in Appendix ??.

C. Proof of converses

The wait-and-overwrite attack runs the same way in both the stochastic and deterministic cases.. The adversary samples $\hat{m}, \tilde{m}$ i.i.d. from the posterior at T_γ , commits to the hypothesis $M = \hat{m}$, and at each step in the overwrite phase applies the matching guaranteed by the channel's overwritability:

- under I-overwritability (deterministic codes), the per-symbol strategy $Q^{(\tau)}$ satisfies

$$\sum_s Q^{(\tau)}(s) W(\cdot | c_{\hat{m}, \tau}, s) = W(\cdot | c_{\tilde{m}, \tau}, s_0);$$

- under distribution-overwritability (stochastic codes), the strategy $Q^{(\tau)}$ matches the encoder's input distribution under $M = \hat{m}$ to the no-adversary distribution under $M = \tilde{m}$ (per-step matching detailed in Appendix ??).

In both cases the adversary chooses the strategy so as to match the output marginal distribution to that under message $M = \tilde{m}$ while assuming that the sent message is $\hat{m}$. The success event for the attack is therefore $\{\hat{m} = M, \tilde{m} \neq M\}$

By Lemma 11, under $\mathcal{E} \triangleq \{T_\gamma \leq n, G_{M|Y^{T_\gamma}} \leq 1/2\}$, we have $\Pr(\mathcal{E}) \geq 3/4 - \epsilon_n/(1 - \sqrt{\gamma})$. Since $\hat{m}, \tilde{m}, M$ are conditionally i.i.d. from the posterior given Y^{T_γ} , the success event has conditional probability $G(1-G)$, which is at least $\gamma(1-\gamma)$ under $\mathcal{E}$. Conditioned on the success event, the channel output is a no-adversary transcript of $\tilde{m} \neq M$, so reliability of the code makes the decoder output $\tilde{m}$ with probability $\geq 1 - \epsilon_n$ – an authentication error. Multiplying gives the bound:

$$\Pr_{\text{adv}}(\hat{M} \notin \{M, \text{REJECT}\}) \geq \gamma(1-\gamma)(1-\epsilon_n) \left[\frac{3}{4} - \frac{\epsilon_n}{1-\sqrt{\gamma}} \right]. \quad (7)$$

Proposition 12 (Converse halves of Theorems 7 and 9). *Fix $\gamma \in (0, 1/(32|\mathcal{Y}|^2))$. For any blocklength- n feedback code (F, ϕ) with auth-error ϵ_n , the wait-and-overwrite attack satisfies (7) provided either*

- (i) (F, ϕ) is a stochastic code and $W_{Y|X,S}$ is distribution-overwritable, or
- (ii) (F, ϕ) is a deterministic code and $W_{Y|X,S}$ is I-overwritable.

In particular, $C_{\text{auth, stoch}}^{\text{fb}} = 0$ in case (i) and $C_{\text{auth, det}}^{\text{fb}} = 0$ in case (ii). This furnishes the converse halves of Theorems 7 and 9.

Remark 13. The two overwritability conditions affect *which* channels admit the attack, not the success probability of the attack. In both cases the adversary's strategy is built around a hypothesis $\hat{m}$ for the true message: under I-OW, because the matching is per-symbol; under dist-OW, because the matching is per encoder input distribution and the encoder's distribution depends on M . The bottleneck is the same event $\{\hat{m} = M\}$, of conditional probability $G \geq \gamma$ on $\mathcal{E}$, giving the $\gamma(1-\gamma)$ factor in (7). The two converses reduce to the same calculation; they differ only in the class of channels to which they apply.

Remark 14 (Encoder feedback does not change the positivity of the capacity). Proposition 12 uses only the encoder's feedback-aware input *distribution*, not the absence of encoder feedback. The same attack applies verbatim with or without encoder feedback, so both cases of the proposition cover C^{fb} and $C^{\text{no-fb}}$.

V. ACHIEVABILITY FOR STOCHASTIC CODES (THEOREMS 7 AND 8)

We outline the achievability proof for stochastic codes. If $W_{Y|X,S}$ is *not* distribution-overwritable, then a stochastic feedback code achieves any rate $R < C_{s_0}$ with vanishing authentication error. The construction combines a good channel code over $W_{Y|X,S=s_0}$ with an *authentication tag*; the key ingredient (Lemma 15) is built from the overlapping-sets lemma of [5] and a martingale concentration argument that handles the sequential nature of the feedback adversary. Full details are in Appendix ??; the deterministic case (Theorem 9) is treated in Section VI as the specialization $P_X = \delta_{x_0}$ of the same template.

A. Authentication tags

An (N, n) -*authentication tag* [5, Def. 9] consists of a stochastic encoder $P_{X^n|M}$ and a family of deterministic verifiers $\{D_{\hat{m}}: \mathcal{Y}^n \rightarrow \{\text{ACCEPT}, \text{REJECT}\}\}_{\hat{m}}$ indexed by a candidate message $\hat{m}$ already available to the receiver, with low false-alarm and missed-detection error (formal definition in Appendix ??).

Lemma 15 (Existence of authentication tags). *If $W_{Y|X,S}$ is not distribution-overwritable, then there exists $R > 0$ such that, for every $(\lambda_1, \lambda_2) \in (0, 1)^2$ and all n sufficiently large, there is an (N_n, n) -tag with errors (λ_1, λ_2) and $\liminf_{n \rightarrow \infty} \frac{1}{n} \log \log N_n \geq R$.*

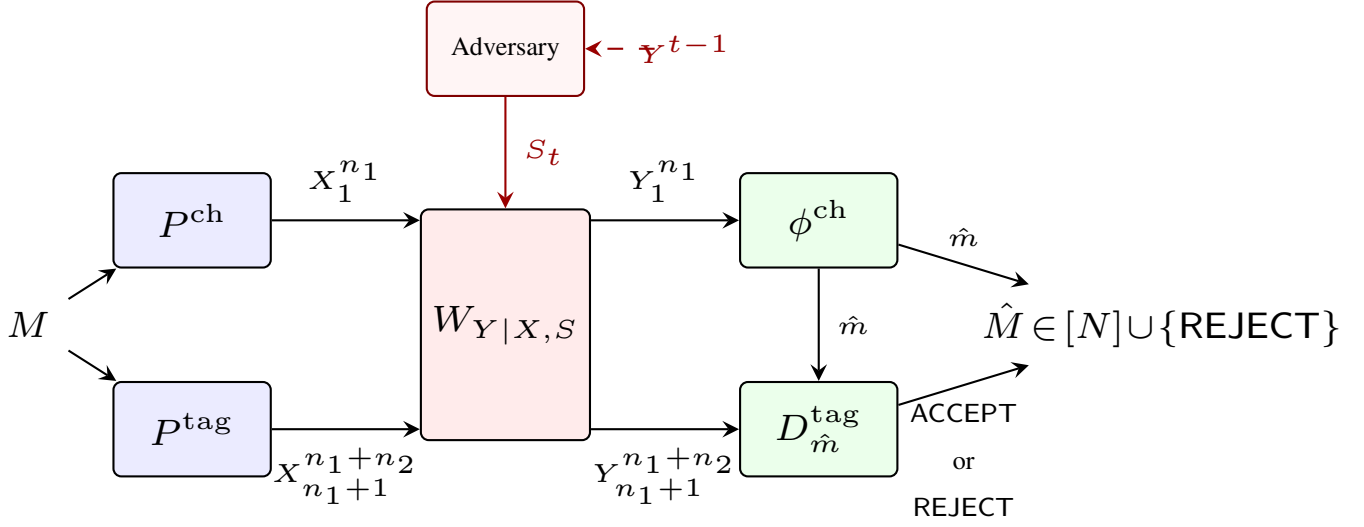


Fig. 2: Achievability for Theorem 8. Channel encoder $P_{X_1^{n_1}|M}^{\text{ch}}$ encodes the first n_1 channel-input symbols from a capacity-achieving code over $W_{Y|X,S=s_0}$; tag encoder $P_{X_{n_1+1}^{n_1+n_2}|M}^{\text{tag}}$ emits the remaining n_2 symbols as an authentication tag for the same M . The decoder splits $Y^{n_1+n_2}$, decodes $\hat{m}$ from the first n_1 outputs, and runs the verifier $D_{\hat{m}}^{\text{tag}}$ on the last n_2 . The output is $\hat{m}$ if the verifier accepts, REJECT otherwise.

Proof sketch: We follow [5, Sec. V]. Fix a non-overwritability witness (P_X, x', δ) from (3) and an *overlapping-sets codebook* $\mathfrak{B} = \{\mathcal{B}_m \subset [n]\}_{m \in [N]}$ [5, Lemma 3]. The encoder sends $X_t = x'$ on $\mathcal{B}_m$ and $X_t \sim P_X$ i.i.d. off $\mathcal{B}_m$; the verifier accepts $\hat{m}$ iff $\mathbb{V}(\hat{P}_{Y^n}(\cdot | \mathcal{B}_{\hat{m}}), W(\cdot | x', s_0)) \leq \delta/4$.

Missed detection ($\hat{m} \neq m$, feedback adversary Q). Let $J \triangleq \mathcal{B}_{\hat{m}} \setminus \mathcal{B}_m$ (the *flip set*). The verifier's TV at $\mathcal{B}_{\hat{m}}$ is controlled by the TV at J via the overlap property [5, Lemma 3.5]), which is purely a set-geometric statement. So it suffices to show

$$\mathbb{V}(\hat{P}_{Y^n}(\cdot | J), W(\cdot | x', s_0)) \geq \delta/2 \quad (8)$$

with high probability. On J the encoder uses P_X , so the per-step output law (conditional on history $\mathcal{F}_{t-1}$) is the averaged channel $\pi_t = \sum_{x,s} P_X(x) Q_{S_t|\mathcal{F}_{t-1}}(s) W(\cdot | x, s)$, which is δ -far from $W(\cdot | x', s_0)$ in TV by the non-overwritability gap (3). The technical step is showing $\hat{P}_{Y^n}(\cdot | J)$ concentrates around the average of π_t : under feedback, Y_t 's couple through Y^{t-1} , so Hoeffding does not apply, but $Z_t \triangleq \mathbf{1}\{Y_t \in A\} - \pi_t(A)$ is a bounded-difference $\mathcal{F}_t$ -martingale, and Azuma–Hoeffding yields (8) with probability $1 - e^{-\Omega(n)}$ uniformly over Q.

This produces only a single-exponential authentication tag $\frac{1}{n} \log N_n^{(\text{auth})} \geq R^{(\text{auth})}$. The doubly-exponential rate claimed in Lemma 15 is obtained by composing it with an Ahlswede–Dueck identification code for the noiseless binary channel; details, again mirroring [5], are in Appendix ??.

B. Authentication codes via composition with a channel code

The achievability scheme uses any capacity-achieving channel code whose decoded output is fed into the authentication-tag verifier as the candidate message.

Fix $R < C_{s_0}$. Choose blocklength n so that:

- (i) there is an $(\lceil 2^{nR} \rceil, n_1)$ channel code $(P_{X^{n_1}|M}^{\text{ch}}, \phi^{\text{ch}})$ over the no-adversary channel $W_{Y|X,S=s_0}$ with maximal error $\epsilon_n = o(1)$;

- (ii) there is an $(\lceil 2^{nR} \rceil, n_2)$ authentication tag $(P_{X^{n_2}|M}^{\text{tag}}, \{D_{\hat{m}}^{\text{tag}}\})$ of error pair (ϵ_n, ϵ_n) for $W_{Y|X,S}$, guaranteed by Lemma 15 as long as $n_2 = \Omega(\log \log \lceil 2^{nR} \rceil) = O(\log n)$.

The composite encoder uses

$$P_{X^{n_1+n_2}|M}(x^{n_1+n_2}|m) = P_{X^{n_1}|M}^{\text{ch}}(x^{n_1}|m) P_{X^{n_2}|M}^{\text{tag}}(x_{n_1+1}^{n_1+n_2}|m).$$

The decoder first runs ϕ^{ch} on Y^{n_1} to obtain a candidate $\hat{m} = \phi^{\text{ch}}(Y^{n_1})$, then verifies it via the tag:

$$\phi(Y^{n_1+n_2}) = \begin{cases} \hat{m} & \text{if } D_{\hat{m}}^{\text{tag}}(Y_{n_1+1}^{n_1+n_2}) = \text{ACCEPT}, \\ \text{REJECT} & \text{otherwise.} \end{cases}$$

Error analysis: Under no adversary, the channel-code stage outputs $\hat{m} = M$ with probability $\geq 1 - \epsilon_n$, and the verifier accepts the correct tag with probability $\geq 1 - \epsilon_n$. So $\Pr_{\text{na}}(\phi \neq M) \leq 2\epsilon_n$.

Under any adversary strategy, the channel-code stage outputs *some* candidate $\hat{m} \in [N]$. If $\hat{m} = M$, the tag verifier accepts and authentication succeeds. If $D_{\hat{m}}^{\text{tag}}(Y_{n_1+1}^{n_1+n_2}) = \text{REJECT}$, the composite decoder outputs REJECT and authentication succeeds. Otherwise $\hat{m} \neq M$ and the verifier accepts – but this is exactly the missed-detection event of the tag, whose probability is at most ϵ_n for every adversary strategy. Combining,

$$\Pr_{\text{adv}}(\phi(Y^{n_1+n_2}) \notin \{M, \text{REJECT}\}) \leq \epsilon_n.$$

Hence the composite code achieves authentication error $\leq 3\epsilon_n = o(1)$ at rate $\frac{nR}{n+O(\log n)} = \frac{nR}{n+O(\log n)} \rightarrow R$ as $n \rightarrow \infty$.

Since $R < C_{s_0}$ was arbitrary, $C_{\text{auth, stoch}}^{\text{fb}} \geq C_{s_0}$, and combined with the trivial reverse inequality this proves Theorem 8. $\square$

Remark 16. A subtlety in the composite scheme is that during the wait phase the adversary may decode M from Y^{n_1} and become effectively omniscient in M during the tag phase, with sequential observations $Y_{n_1+1}, Y_{n_1+2}, \dots$. The tag's missed-detection bound is nevertheless uniform over all such adversaries. The encoder's fresh i.i.d. P_X randomness on the flip set $\mathcal{B}_{\hat{m}} \setminus \mathcal{B}_m$ hides X_t at each individual position even from an M -omniscient adversary, and the non-distribution-overwritability gap (3) bounds the conditional output law π_t uniformly over the adversary's mixing $Q_{S_t|\mathcal{F}_{t-1}}$. The flip-set martingale closes the bound regardless of what wait-phase information the adversary has accumulated. Details are in Appendix ??.

VI. ACHIEVABILITY FOR DETERMINISTIC CODES (THEOREM 9)

We prove the achievability direction of Theorem 9 as the specialization $P_X = \delta_{x_0}$ of the stochastic construction of Section V; if $W_{Y|X,S}$ is not I-overwritable then $C_{\text{auth, det}}^{\bullet} > 0$. Concretely, the witness (P_X, x', δ) becomes $(\delta_{x_0}, x'_0, \delta)$, with δ now the *I-overwritability* gap of (4). The key differences from the stochastic case are in three places:

- *Encoder.* With $P_X = \delta_{x_0}$, the encoder is fully deterministic: $X_t = x'_0$ for $t \in \mathcal{B}_m$ and $X_t = x_0$ for $t \notin \mathcal{B}_m$. The codeword is binary, $c_m \in \{x_0, x'_0\}^n$.
- *Witness gap.* The dist-OW gap in step (1) of the auth-tag proof reduces, since P_X is a Dirac mass, to the I-OW gap (4), namely

$$\inf_Q \mathbb{V}(\sum_s Q(s) W(\cdot | x_0, s), W(\cdot | x'_0, s_0)) \geq \delta.$$

- *Step (2) martingale.* On the flip set the encoder transmits x_0 deterministically; the conditional Bernoulli mean is $p_t = \sum_s Q_{S_t|\mathcal{F}_{t-1}}(s | \mathcal{F}_{t-1}) W(A | x_0, s)$, δ -far from $W(A | x'_0, s_0)$ uniformly over Q . Azuma–Hoeffding on the same bounded-difference martingale gives the same flip-set concentration.

The codebook (overlapping-sets lemma) and the overlap property (step (3)) are unchanged – they are statements about set geometry and do not depend on the input distribution. Combining gives $C_{\text{auth,det}}^\bullet \geq R > 0$ for both $\bullet \in \{\text{fb}, \text{no-fb}\}$. The full deterministic proof is in Appendix ??.

Remark 17. The proof yields a positive rate strictly below the Shannon capacity C^* of the no-adversary sub-channel $W^*(y|x) \triangleq W(y|x, s_0)$ on $\{x_0, x'_0\}$. The gap to C^* (and a fortiori to C_{s_0}) is an open problem – see Remark 10.

VII. EXAMPLES AND THE STRICT HIERARCHY

We close with two channels that together exhibit the strict hierarchy oblivious-OW $\subsetneq$ distribution-OW $\subsetneq$ I-OW.

Example 18 (I-overwritable but not distribution-overwritable). Let $\mathcal{X} = \mathcal{Y} = \{0, 1\}$, $\mathcal{S} = \{s_0, s_1\}$, with $W_{Y|X,S}(y|x, s_0) = \mathbf{1}\{y=x\}$ and $W_{Y|X,S}(y|x, s_1) = \mathbf{1}\{y \neq x\}$. For any $(x, x') \in \mathcal{X}^2$, the strategy $Q(s_0) = \mathbf{1}\{x=x'\}$, $Q(s_1) = \mathbf{1}\{x \neq x'\}$ matches $W_{Y|X,S}(\cdot|x', s_0)$, so $W_{Y|X,S}$ is I-overwritable. It is not distribution-overwritable: for target $x' = 0$ and $P_X(0) = p \in (0, 1)$, an oblivious mixing $Q(s_0) = q$ gives $\Pr(Y=0) = pq + (1-p)(1-q) = 1$ only at $\{p, q\} \in \{(0, 1), (1, 0)\}$, contradicting $p \in (0, 1)$. Hence $C_{\text{auth,det}}^\bullet = 0$ (Theorem 9) but $C_{\text{auth,stoch}}^\bullet = C_{s_0} = 1$ bit/use (Theorems 7–8).

Example 19 (Distribution-overwritable but not obviously overwritable). The AZBSC(δ, α), adapted from [5], has $\mathcal{X} = \mathcal{Y} = \{0, 1\}$, $\mathcal{S} = \{s_0, s_{1,0}, s_{1,1}\}$, and parameters $0 < \alpha < \delta < 1/2$: $W_{Y|X,S}(\cdot|x, s_0)$ is a BSC(δ); $W_{Y|X,S}(y|x, s_{1,x}) = \mathbf{1}\{y=x\}$; and $W_{Y|X,S}(y|x, s_{1,x \oplus 1}) = \alpha \mathbf{1}\{y=x\} + (1-\alpha) \mathbf{1}\{y \neq x\}$. Distribution-overwritability holds: for $P_X(0) = p$ and target $x' = 0$, the strategy $Q(s_{1,0}) = \frac{1-\delta-p\alpha}{1-\alpha}$, $Q(s_{1,1}) = \frac{\delta-(1-p)\alpha}{1-\alpha}$ matches $W_{Y|X,S}(\cdot|0, s_0)$. Oblivious overwritability fails because matching for both $x = 0$ and $x = 1$ forces $q_0 + q_1 = (1-2\delta)/(1-2\delta-\alpha) \notin [0, 1]$ for any admissible (δ, α) . Hence $C_{\text{auth,stoch}}^\bullet = C_{s_0}$ by Theorems 7–8; full verification is in Appendix ??.

Together, Examples 18 and 19 show that all three overwritability classes are genuinely distinct, and that Theorems 7 and 9 each give a strict tightening of the converse.

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TABLE II: Proof roadmap. Building blocks for the converse and achievability are listed first; the main theorems combine the two sides.

Result	Proved in	Uses
<i>Converse-side building blocks</i>		
Lemma ?? (sub-multiplicativity)	App. ??	—
Lemma 11 (stopping time)	App. ??	Lemma ??
Claim ?? (per-step matching)	App. ??	dist-OW (Def. 3) / I-OW (Def. 2)
Proposition 12 (converse halves)	App. ??	Lemma 11, Claim ??
<i>Achievability-side building blocks</i>		
Lemma 15 (auth-tag existence)	App. ??, ??	non-distribution-OW gap (3), identification codes
Composite encoder analysis	App. ??	Lemma 15
<i>Main theorems (combining both sides)</i>		
Theorem 7 (stochastic positivity)	Achiev.: §V, App. ??; Conv.: App. ??, case (i)	Lemma 15, Proposition 12
Theorem 8 (stochastic capacity)	Achiev.: §V, App. ??; Conv.: App. ??	Lemma 15, Proposition 12
Theorem 9 (deterministic positivity)	Achiev.: §VI, App. ??; Conv.: App. ??, case (ii)	Const.-comp. codebook + I-OW, Proposition 12

APPENDIX A

SUB-MULTIPLICATIVE GROWTH OF THE GUESSING PROBABILITY

Lemma 20 (Guessing-probability sub-multiplicativity). *Under the no-adversary law na,*

$$\mathbb{E}[\sqrt{G_{M|Y^t, S^t}(Y^t, s_0^t)} \mid Y^{t-1} = y^{t-1}] \leq |\mathcal{Y}| \sqrt{G_{M|Y^{t-1}, S^{t-1}}(y^{t-1}, s_0^{t-1})}.$$

Proof. Write $g(y^t) \triangleq G_{M|Y^t, S^t}(y^t, s_0^t)$ and $p(y_t | y^{t-1}) \triangleq P_{Y_t|Y^{t-1}, S^t}(y_t | y^{t-1}, s_0^t)$. For every (m, y_t) ,

$$P_{M, Y_t|Y^{t-1}}(m, y_t | y^{t-1}) \leq P_{M|Y^{t-1}}(m | y^{t-1}).$$

Squaring and summing over m gives

$$\sum_m P_{M, Y_t|Y^{t-1}}(m, y_t | y^{t-1})^2 \leq g(y^{t-1}).$$

Therefore

$$\begin{aligned} \mathbb{E}[\sqrt{g(Y^t)} | y^{t-1}] &= \sum_{y_t} p(y_t | y^{t-1}) \sqrt{\sum_m P_{M|Y^t}(m | y^t)^2} \\ &= \sum_{y_t} \sqrt{\sum_m (p(y_t | y^{t-1}) P_{M|Y^t}(m | y^t))^2} = \sum_{y_t} \sqrt{\sum_m P_{M, Y_t|Y^{t-1}}(m, y_t | y^{t-1})^2} \leq \sum_{y_t} \sqrt{g(y^{t-1})} = |\mathcal{Y}| \sqrt{g(y^{t-1})}. \end{aligned}$$

□

APPENDIX B

PROOF OF LEMMA 11

Throughout this appendix all probabilities and expectations are taken under the no-adversary law na , i.e., the joint law of (M, X^n, Y^n) when the adversary plays $S^n = s_0^n$. Write $g(y^t) \triangleq G_{M|Y^t, S^t}(y^t, s_0^t)$ and recall $T_\gamma = \inf \{t \in [n] : g(Y^t) > \gamma\}$.

Proof of (5). Let $p_e^{(n)} \triangleq \Pr_{\text{na}}(\hat{M} \neq M)$, and write $p^*(y^n) \triangleq 1 - \max_m P_{M|Y^n}(m|y^n)$ for the conditional MAP error given $Y^n = y^n$. Since the MAP rule is optimal, $\mathbb{E}[p^*(Y^n)] \leq p_e^{(n)}$. The guessing probability dominates the squared MAP success probability:

$$g(y^n) \geq \left(\max_m P_{M|Y^n}(m|y^n) \right)^2 = (1 - p^*(y^n))^2,$$

so $g(Y^n) \leq \gamma$ forces $p^*(Y^n) \geq 1 - \sqrt{\gamma}$. By Markov,

$$\Pr(T_\gamma > n) \leq \Pr(g(Y^n) \leq \gamma) \leq \frac{p_e^{(n)}}{1 - \sqrt{\gamma}} \xrightarrow{n \rightarrow \infty} 0.$$

Proof of (6). Under the event $\{T_\gamma = t\}$, the definition of T_γ as the *first* crossing time gives $g(Y^{t-1}) \leq \gamma$. Combining with Lemma ??,

$$\mathbb{E}[\sqrt{g(Y^t)} \mid T_\gamma = t] \leq |\mathcal{Y}| \sqrt{\gamma}.$$

Markov's inequality with $\gamma < 1/(32|\mathcal{Y}|^2)$ yields

$$\begin{aligned} \Pr(g(Y^t) > \tfrac{1}{2} \mid T_\gamma = t) &\leq \sqrt{2} \cdot \mathbb{E}[\sqrt{g(Y^t)} \mid T_\gamma = t] \\ &\leq \sqrt{2} |\mathcal{Y}| \sqrt{\gamma} < \tfrac{1}{4}. \end{aligned}$$

Averaging over $t \in [n]$ weighted by $\Pr(T_\gamma = t)$ gives $\Pr(g(Y^{T_\gamma}) > 1/2 \mid T_\gamma \leq n) < 1/4$, i.e., (6). $\square$

APPENDIX C

DETAILED PROOF OF THE CONVERSE

We give a single detailed proof of the converse Proposition 12, covering both the stochastic/distribution-overwritable and the deterministic/I-overwritable cases. We first establish the per-step matching identity (Claim ??); then write the attack joint distribution explicitly; decompose the resulting distribution into a “matched” part ($M = \hat{M}$) and a “mismatched” part ($M \neq \hat{M}$); bound the rejection probability and the correct-decoding probability separately; and finally combine via the high-probability event $\mathcal{E}$.

Throughout, (F, ϕ) is a feedback code of blocklength n with auth-error ϵ_n , $\mathcal{D}_\perp \triangleq \phi^{-1}(\text{REJECT})$, and the wait-and-overwrite attack of Section IV stops at T_γ and overwrites with $\tilde{M}$. We use $\tilde{\mathbf{P}}$ for the joint distribution under the attack and $\mathbf{P}_{\text{na}}$ for the no-adversary joint. The argument is written for the stochastic case; the deterministic case is the specialization in which $\tilde{\mathbf{P}}^{(\tau)} = \mathbf{1}_{c_{\hat{m}, \tau}}$ is a point mass and dist-OW reduces to I-OW (cf. end of this appendix).

Per-step matching:

Claim 21. Suppose $W_{Y|X, S}$ is distribution-overwritable. Fix $\tau \in [T_\gamma + 1 : n]$ and a history $h^{\tau-1} \triangleq (\hat{m}, \tilde{m}, y^{\tau-1}, s^{\tau-1})$. Let $\tilde{\mathbf{P}}^{(\tau)}$ be the encoder's marginal at step τ under the attack with $M = \hat{M} = \hat{m}$, and let $\mathbf{P}^{(\tau)}$ be the encoder's marginal at step τ under no-adversary with $M = \tilde{m}$. Then there exists $\mathbf{Q}^{(\tau)} \in \mathcal{P}(\mathcal{S})$ with

$$\sum_{x_\tau, s_\tau} \tilde{\mathbf{P}}^{(\tau)}(x_\tau) \mathbf{Q}^{(\tau)}(s_\tau) W_{Y|X, S}(y_\tau | x_\tau, s_\tau) = \sum_{x'_\tau} \mathbf{P}^{(\tau)}(x'_\tau) W_{Y|X, S}(y_\tau | x'_\tau, s_0)$$

for every $y_\tau \in \mathcal{Y}$.

Proof. $\tilde{P}^{(\tau)}$ is computable from the encoder's conditional law and the realized $(\hat{m}, y^{\tau-1}, s^{\tau-1})$ via the path likelihood

$$P_{X_\tau|M, Y^{\tau-1}, S^{\tau-1}}(x_\tau | m, h^{\tau-1}) = \frac{\Lambda_\tau(m, x_\tau, h^{\tau-1})}{\sum_{x'_\tau} \Lambda_\tau(m, x'_\tau, h^{\tau-1})},$$

where

$$\Lambda_\tau(m, x_\tau, h^{\tau-1}) = \sum_{x^{\tau-1}} \left(\prod_{i=1}^{\tau-1} P_{X_i|M, X^{i-1}, Y^{i-1}}(x_i | m, x^{i-1}, y^{i-1}) \times W_{Y|X, S}(y_i | x_i, s_i) \right) \times P_{X_\tau|M, X^{\tau-1}, Y^{\tau-1}}(x_\tau | m, x^{\tau-1}, y^{\tau-1}).$$

Apply Remark 4 with overwritten distribution $P = \tilde{P}^{(\tau)}$ and target $P' = P^{(\tau)}$ to obtain $Q^{(\tau)}$ satisfying (??). $\square$

Telescoping: For $i \in [T_\gamma + 1 : n]$, define the hybrid trajectory

$$\Gamma_i(y_{T_\gamma+1}^n | \hat{m}, \tilde{m}, y^{T_\gamma}) \triangleq A_i \cdot B_i,$$

where

$$A_i \triangleq \sum_{s_{T_\gamma+1}^i} \prod_{\tau=T_\gamma+1}^i \tilde{P}_{Y_\tau, S_\tau|M, \hat{M}, \tilde{M}, Y^{\tau-1}, S^{\tau-1}}(y_\tau, s_\tau | \dots),$$

$$B_i \triangleq \prod_{\tau=i+1}^n \sum_{x_\tau} P_{X_\tau|M, Y^{\tau-1}, S^{\tau-1}}(x_\tau | \tilde{m}, y^{\tau-1}, s_0^{\tau-1}) \times W_{Y|X, S}(y_\tau | x_\tau, s_0).$$

A_i records the actual attack up through step i , and B_i the no-adversary distribution thereafter. Substituting (??) into the inner sum over s_i inside A_i collapses the $\tau=i$ factor of A_i onto the $\tau=i$ factor of B_{i-1} , so $\Gamma_i = \Gamma_{i-1}$. Iterating from $i=n$ down to $i=T_\gamma + 2$ yields the telescoping identity

$$\Gamma_n = \Gamma_{T_\gamma+1}.$$

Explicit attack joint distribution:

For $t \in [n]$ and $y^n \in \mathcal{G}_t = \{T_\gamma(y^n) = t\}$, the wait phase forces $S^t = s_0^t$ and the overwrite phase samples $\hat{M}, \tilde{M}$ i.i.d. from $P_{M|Y^t, S^t}^{\text{na}}(\cdot | y^t, s_0^t)$ and uses the strategies $\{Q^{(\tau)}\}_{\tau > t}$. Hence

$$\tilde{P}(M=m, \hat{M}=\hat{m}, \tilde{M}=\tilde{m}, Y^n=y^n, S^n=(s_0^t, s_{t+1}^n)) = P_M(m) P_{Y^t|M, S^t}^{\text{na}}(y^t | m, s_0^t) P_{M|Y^t, S^t}^{\text{na}}(\hat{m} | y^t, s_0^t) \cdot P_{M|Y^t, S^t}^{\text{na}}(\tilde{m} | y^t, s_0^t) \cdot \tilde{P}_{Y_{t+1}^n, S_{t+1}^n}$$

Term-by-term decomposition:

Marginalize (??) over $(\hat{m}, m, s_{t+1}^n)$ and split on $m = \hat{m}$ vs $m \neq \hat{m}$:

$$\tilde{P}(Y^n = y^n) = P_{Y^t|S^t}^{\text{na}}(y^t | s_0^t) [A_{\text{term}}(y^n) + B_{\text{term}}(y^n)],$$

where

$$A_{\text{term}}(y^n) \triangleq \sum_{\hat{m}, \tilde{m}} (P_{M|Y^t, S^t}^{\text{na}}(\hat{m} | y^t, s_0^t))^2 \cdot P_{M|Y^t, S^t}^{\text{na}}(\tilde{m} | y^t, s_0^t) \Gamma_n,$$

$$B_{\text{term}}(y^n) \triangleq \sum_{\substack{m, \tilde{m} \\ \hat{m} \neq m}} P_{M|Y^t, S^t}^{\text{na}}(m | \cdot) P_{M|Y^t, S^t}^{\text{na}}(\hat{m} | \cdot) \cdot P_{M|Y^t, S^t}^{\text{na}}(\tilde{m} | \cdot) \Gamma_n.$$

By the telescoping identity (??), $\Gamma_n = \Gamma_{T_\gamma+1}$, which is independent of $\hat{m}$ and equals the no-adversary likelihood of Y_{t+1}^n given $M = \tilde{m}$ and $Y^t = y^t$. Substituting into A_{term} and using $\sum_{\hat{m}} (P_{M|Y^t, S^t}^{\text{na}}(\hat{m} | \cdot))^2 = G_{M|Y^t, S^t}(y^t, s_0^t)$,

$$A_{\text{term}}(y^n) = G_{M|Y^t, S^t}(y^t, s_0^t) \times P_{Y_{t+1}^n|Y^t, S^n}^{\text{na}}(y_{t+1}^n | y^t, s_0^n).$$

Multiplying by $P_{Y^t|S^t}^{\text{na}}(y^t | s_0^t)$ and using $G_{M|Y^t, S^t} \leq 1$,

$$P_{Y^t|S^t}^{\text{na}}(y^t | s_0^t) A_{\text{term}}(y^n) \leq \Pr_{\text{na}}(Y^n = y^n).$$

For B_{term} , marginalize over $\tilde{m}$ first ($\sum_{\tilde{m}, y_{t+1}^n} P_{M|Y^t, S^t}^{\text{na}} \Gamma_n = 1$); then $\sum_{\tilde{m} \neq m} P_{M|Y^t, S^t}^{\text{na}}(\tilde{m} | \cdot) = 1 - P_{M|Y^t, S^t}^{\text{na}}(m | \cdot)$, and $\sum_m P_{M|Y^t, S^t}^{\text{na}}(m | \cdot)(1 - P_{M|Y^t, S^t}^{\text{na}}(m | \cdot)) = 1 - G_{M|Y^t, S^t}(y^t, s_0^t)$. Hence

$$\sum_{y_{t+1}^n} B_{\text{term}}(y^n) = 1 - G_{M|Y^t, S^t}(y^t, s_0^t) \leq 1 - \gamma \quad \text{for } y^t \in \mathcal{F}_t,$$

by the stopping condition $G_{M|Y^t, S^t} > \gamma$ on $\mathcal{F}_t$.

Total bound on $\tilde{P}(\mathcal{D}_\perp)$:

Combining (??) and (??) and summing over t ,

$$\begin{aligned} \tilde{P}(\mathcal{D}_\perp) &= \sum_{t=1}^n \sum_{y^n \in \mathcal{G}_t \cap \mathcal{D}_\perp} \tilde{P}(Y^n = y^n) \\ &\leq \Pr_{\text{na}}(\mathcal{D}_\perp) + (1 - \gamma) \sum_t \Pr_{\text{na}}(T_\gamma = t) + \Pr_{\text{na}}(T_\gamma > n) \\ &\leq \epsilon_n + (1 - \gamma) + \frac{\epsilon_n}{1 - \sqrt{\gamma}}, \end{aligned}$$

where the last step uses Lemma 11(5) for the residual mass on $\{T_\gamma > n\}$ and $\Pr_{\text{na}}(\mathcal{D}_\perp) \leq \epsilon_n$.

Total bound on $\tilde{P}(\hat{M} = M)$:

Repeat the decomposition above but now sum y^n over the event $\{\phi(y^n) = m\}$ rather than $\mathcal{D}_\perp$. After using the same telescoping identity (i.e., (??) absorbed into a no-adversary likelihood for message $\tilde{m}$), and splitting on $m = \tilde{m}$ vs $m \neq \tilde{m}$,

$$\begin{aligned} \tilde{P}(\hat{M} = M) &\leq \sum_{m, t} P_M(m) P_{M|Y^t, S^t}^{\text{na}}(m | y^t, s_0^t) + \epsilon_n \\ &\leq \mathbb{E}_{\text{na}}[G_{M|Y^t, S^t}(Y^{T_\gamma}, s_0^{T_\gamma})] + \epsilon_n. \end{aligned}$$

By Lemma 11(6), $G_{M|Y^t, S^t} \leq 1/2$ with conditional probability $\geq 3/4$ on $\{T_\gamma \leq n\}$ and $G_{M|Y^t, S^t} \leq 1$ otherwise; together with (5),

$$\mathbb{E}_{\text{na}}[G_{M|Y^t, S^t}(Y^{T_\gamma}, s_0^{T_\gamma})] \leq \frac{5}{8} + \frac{\epsilon_n}{1 - \sqrt{\gamma}}.$$

Combining:

The events $\{\hat{M} = \text{REJECT}\}$ and $\{\hat{M} = M\}$ are disjoint, so

$$\tilde{P}(\hat{M} \notin \{M, \text{REJECT}\}) = 1 - \tilde{P}(\hat{M} = \text{REJECT}) - \tilde{P}(\hat{M} = M) \geq \gamma - \frac{5}{8} - 2\epsilon_n - \frac{2\epsilon_n}{1 - \sqrt{\gamma}}.$$

Bound (??) is loose because γ is small. The tighter bound used in Proposition 12 comes from conditioning on the high-probability event $\mathcal{E} = \{T_\gamma \leq n, G_{M|Y^t, S^t} \leq 1/2\}$. Since $M, \hat{M}, \tilde{M}$ are conditionally i.i.d. from the posterior given Y^{T_γ} , and since $x \mapsto x(1 - x)$ is increasing on $[0, 1/2]$,

$$\tilde{P}(\hat{M} = M, \tilde{M} \neq M | \mathcal{E}) = G_{M|Y^t, S^t}(1 - G_{M|Y^t, S^t}) \geq \gamma(1 - \gamma).$$

On $\mathcal{E} \cap \{\hat{M} = M, \tilde{M} \neq M\}$ the matching of Claim ?? (or, in the deterministic case, the I-OW per-symbol match) makes the channel output a no-adversary transcript of $\tilde{M}$, and reliability gives $\Pr_{\text{na}}(\phi(Y^n) = \tilde{M} | M = \tilde{M}) \geq 1 - \epsilon_n$, so the decoder produces an authentication error with conditional probability $\geq \gamma(1 - \gamma)(1 - \epsilon_n)$ on $\mathcal{E}$. By Lemma 11, $\Pr_{\text{adv}}(\mathcal{E}) \geq 3/4 - \epsilon_n/(1 - \sqrt{\gamma})$.

Multiplying,

$$\tilde{P}(\hat{M} \notin \{M, \text{REJECT}\}) \geq \gamma(1 - \gamma)(1 - \epsilon_n) \left[\frac{3}{4} - \frac{\epsilon_n}{1 - \sqrt{\gamma}} \right],$$

which is (7). □

Specialization to the deterministic case:

For deterministic codes, the encoder's marginal $\tilde{P}^{(\tau)}$ at step τ is a point mass at $c_{\hat{m},\tau}$. The matching condition (??) then reduces to the per-symbol identity $\sum_s Q^{(\tau)}(s)W(\cdot | c_{\hat{m},\tau}, s) = W(\cdot | c_{\hat{m},\tau}, s_0)$, which is exactly I-overwritability for the pair $(c_{\hat{m},\tau}, c_{\hat{m},\tau})$. The rest of the proof is identical, and the same bound (??) holds.

Encoder feedback does not change the proof:

The matching at step τ uses $\tilde{P}^{(\tau)} = \tilde{P}_{X_\tau | M=\hat{m}, \hat{m}, Y^{\tau-1}, S^{\tau-1}}$, which the adversary can compute from $(\hat{m}, y^{\tau-1}, s^{\tau-1})$. A feedback encoder that, having access to $Y^{\tau-1}$ as well, makes X_τ depend on the past does not change this — the adversary, having access to $Y^{\tau-1}$ as well, can compute the encoder's distribution exactly. Hence the same proof covers both C^{fb} and $C^{\text{no-fb}}$.

Remark 22 (Sharpness). The constant $\frac{3\gamma(1-\gamma)}{4}$ is an artifact of Markov's inequality at threshold $1/2$ in Lemma 11 and the loose bound $G(1-G) \geq \gamma(1-\gamma)$ on $\mathcal{E}$. Iterating the stopping rule, using a tighter Doob argument on $\sqrt{G(Y^t, s_0^t)}$, or accounting for the actual joint $G - \sum_m P(m)^3$ can improve the constant; the qualitative conclusion is unchanged.

APPENDIX D

DETAILED PROOF OF STOCHASTIC ACHIEVABILITY (THEOREMS 7 AND 8)

Definition 23 ((N, n) -Authentication tag [5, Def. 9]). An (N, n) -authentication tag for $W_{Y|X,S}$ consists of a stochastic encoder $P_{X^n|M}: [N] \rightarrow \mathcal{P}(\mathcal{X}^n)$ together with a family of deterministic verifiers $\{D_{\hat{m}}: \mathcal{Y}^n \rightarrow \{\text{ACCEPT}, \text{REJECT}\}\}_{\hat{m} \in [N]}$, indexed by a candidate $\hat{m} \in [N]$ already available to the receiver. The tag achieves error pair (λ_1, λ_2) if

$$\begin{aligned} \max_{m \in [N]} \Pr(D_m(Y^n) = \text{REJECT} \mid M=m, S^n=s_0^n) &\leq \lambda_1, \\ \max_{\substack{m \in [N] \\ \hat{m} \neq m}} \sup_Q \Pr(D_{\hat{m}}(Y^n) = \text{ACCEPT} \mid M=m) &\leq \lambda_2, \end{aligned}$$

where Q ranges over feedback-aware adversary strategies. The *tag capacity* C_{tag} is the supremum over R such that for every $(\lambda_1, \lambda_2) \in (0, 1)^2$ there is a sequence of (N_n, n) -tags achieving (λ_1, λ_2) with $\liminf_n \frac{1}{n} \log \log N_n \geq R$.

A. Proof of Lemma 15

The proof has two parts. The first part constructs an *authentication code* of single-exponential rate from the overlapping-sets lemma of [5] and a flip-set martingale concentration; the second part lifts this auth-code to the doubly-exponential rate claimed in Lemma 15 by composing with an Ahlswede–Dueck identification code.

Auth-code construction (single-exponential rate):

We use the same code construction as in [5, Sec. V], based on the overlapping-sets lemma [5, Lemma 3]. The construction is a single template that handles both the deterministic and the stochastic cases through one parameter setting; we state it in stochastic form (general witness distribution P_X). The deterministic case (Section VI) is the specialization $P_X = \delta_{x_0}$ called out at the end of this proof.

Non-overwritability witness: Since $W_{Y|X,S}$ is not distribution-overwritable, by Remark 6 (Eq. (3)) there exist $P_X \in \mathcal{P}(\mathcal{X})$, $x' \in \mathcal{X}$, and $\delta > 0$ with

$$\kappa(P_X, x') \triangleq \min_{Q \in \mathcal{P}(\mathcal{S})} \mathbb{V}(\sum_{x,s} P_X(x) Q(s) W(\cdot | x, s), W(\cdot | x', s_0)) \geq \delta.$$

The pair (P_X, x') is the *non-overwritability witness*: P_X is the encoder's witness input distribution off the verification set, and x' is the target symbol against which the verifier compares.

Codebook (overlapping-sets lemma): By [5, Lemma 3] (which builds on the general Gilbert bound for constant-composition codes, [32, Problem 10.1]), for any $\alpha, \beta \in (0, 1)$ and any rate $R < \min\{\beta^2\alpha^2/6, \beta^2\alpha(1-\alpha)/4\}$ there exists a family $\mathfrak{B} = \{\mathcal{B}_m \subseteq [n] : m \in [N]\}$ with $N \geq 2^{Rn}$ satisfying:

- (i) $\alpha(1-\beta)n \leq |\mathcal{B}_m| \leq \alpha(1+\beta)n$ for all m ;
- (ii) $|\mathcal{B}_m \cap \mathcal{B}_{\hat{m}}| < \alpha^2(1+\beta)n$ for all $m \neq \hat{m}$;
- (iii) $|\mathcal{B}_{\hat{m}} \setminus \mathcal{B}_m| > \alpha(1-\alpha)(1+\beta)n$ for all $m \neq \hat{m}$;
- (iv) (*Overlap property*.) For every $y^n \in \mathcal{Y}^n$ and every $m, \hat{m}$,

$$\mathbb{V}(\hat{P}_{y^n}(\cdot | \mathcal{B}_{\hat{m}}), W(\cdot | x', s_0)) \geq \frac{(1-\alpha)(1-\beta)}{1+\beta} \mathbb{V}(\hat{P}_{y^n}(\cdot | \mathcal{B}_{\hat{m}} \setminus \mathcal{B}_m), W(\cdot | x', s_0)) - \frac{2\alpha(1+\beta)}{1-\beta},$$

where $\hat{P}_{y^n}(\cdot | \mathcal{B})$ is the empirical type of y^n on positions $t \in \mathcal{B}$. Property (iv) is what handles the match-vs-flip cancellation in the analysis below: the TV at the full verification set is bounded below by the TV at the flip set $\mathcal{B}_{\hat{m}} \setminus \mathcal{B}_m$ alone, modulo a small additive correction tunable through α, β .

Encoder: Conditioned on $M = m$:

$$X_t = \begin{cases} x' & \text{if } t \in \mathcal{B}_m, \\ \sim P_X \text{ i.i.d.} & \text{if } t \notin \mathcal{B}_m, \end{cases}$$

i.e., on the message-specific subset $\mathcal{B}_m$ the encoder transmits x' deterministically, and off that subset it draws i.i.d. from the witness distribution P_X .

Verifier: Given candidate $\hat{m}$ and received Y^n , accept iff

$$\mathbb{V}(\hat{P}_{Y^n}(\cdot | \mathcal{B}_{\hat{m}}), W(\cdot | x', s_0)) \leq \delta/4.$$

Choice of α, β : Pick $\alpha, \beta \in (0, 1)$ small enough that

$$\frac{(1-\alpha)(1-\beta)}{1+\beta} \cdot \frac{\delta}{2} - \frac{2\alpha(1+\beta)}{1-\beta} > \frac{\delta}{4}.$$

This is achievable since the LHS tends to $\delta/2 > \delta/4$ as $\alpha, \beta \rightarrow 0$.

No-adversary analysis (false alarm): Fix $m = \hat{m}$ and $S^n = s_0^n$. For $t \in \mathcal{B}_m$ the encoder transmits x' deterministically and $Y_t \sim W(\cdot | x', s_0)$ i.i.d. By Hoeffding plus a $|\mathcal{Y}|$ -fold union bound,

$$\Pr(\mathbb{V}(\hat{P}_{Y^n}(\cdot | \mathcal{B}_m), W(\cdot | x', s_0)) > \delta/4) \leq 2|\mathcal{Y}|e^{-c_1 n}$$

for some $c_1 > 0$ depending only on $\alpha, \delta, |\mathcal{Y}|$. Union-bounded over $N \leq 2^{nR}$ messages, the false-alarm error is $o(1)$ provided $R \ln 2 < c_1$.

Adversarial analysis (missed detection): Fix $m \neq \hat{m}$ and a feedback-aware adversary $Q = \{Q_{S_t | S^{t-1}, Y^{t-1}}\}$. By the overlap property (??) and the choice of α, β in (??), acceptance of $\hat{m}$ (i.e., (??)) implies

$$\mathbb{V}(\hat{P}_{Y^n}(\cdot | \mathcal{B}_{\hat{m}} \setminus \mathcal{B}_m), W(\cdot | x', s_0)) < \delta/2.$$

We now bound the flip-set TV explicitly. Set $J_{\text{flip}} \triangleq \mathcal{B}_{\hat{m}} \setminus \mathcal{B}_m$ and let $N \triangleq |J_{\text{flip}}| > \alpha(1-\alpha)(1+\beta)n$ by (iii). Order the flip positions as $t_1 < t_2 < \dots < t_N$.

The natural filtration adapted to the sequential adversary is

$$\mathcal{F}_t \triangleq \sigma(M, \hat{M}, X^t, S^t, Y^t).$$

On the flip set, $t \notin \mathcal{B}_m$, so the encoder transmits $X_t \sim P_X$ i.i.d., conditionally independent of $\mathcal{F}_{t-1}$. The adversary then picks S_t from the kernel $Q_{S_t|\mathcal{F}_{t-1}}(\cdot | \mathcal{F}_{t-1})$, and the channel output is $Y_t \sim W(\cdot | X_t, S_t)$. For any test set $A \subseteq \mathcal{Y}$ and any $t \in J_{\text{flip}}$,

$$\Pr(Y_t \in A | \mathcal{F}_{t-1}) = \sum_{x,s} P_X(x) Q_{S_t|\mathcal{F}_{t-1}}(s | \mathcal{F}_{t-1}) W(A | x, s) =: p_t^{(A)}.$$

The kernel $Q_{S_t|\mathcal{F}_{t-1}}$ encodes the adversary's sequential strategy. Note that $p_t^{(A)}$ is $\mathcal{F}_{t-1}$ -measurable. By the non-overwritability gap (??),

$$|p_t^{(A)} - W(A | x', s_0)| \geq \delta \quad \text{a.s., uniformly over } Q.$$

We define the following bounded-difference martingale. Let

$$Z_t^{(A)} \triangleq \mathbf{1}\{Y_t \in A\} - p_t^{(A)}, \quad t \in J_{\text{flip}},$$

and the partial sum $M_T^{(A)} \triangleq \sum_{k=1}^T Z_{t_k}^{(A)}$. Since $\mathbb{E}[Z_t^{(A)} | \mathcal{F}_{t-1}] = 0$ a.s. and $|Z_t^{(A)}| \leq 1$, $\{M_T^{(A)}\}$ is an $\{\mathcal{F}_{t_T}\}$ -adapted martingale with bounded differences. Azuma–Hoeffding gives, for any $\xi > 0$,

$$\Pr(|M_N^{(A)}| > N\xi) \leq 2e^{-N\xi^2/2}.$$

Next, let $\hat{P}_{Y^n}(A | J_{\text{flip}}) = \frac{1}{N} \sum_{k=1}^N \mathbf{1}\{Y_{t_k} \in A\}$ and $\bar{p}^{(A)} \triangleq \frac{1}{N} \sum_{k=1}^N p_{t_k}^{(A)}$, (??) reads

$$\Pr(|\hat{P}_{Y^n}(A | J_{\text{flip}}) - \bar{p}^{(A)}| > \xi) \leq 2e^{-N\xi^2/2}.$$

Choose $A = A^*$ to be the TV-attaining set in (??); then (??) averaged over $t \in J_{\text{flip}}$ gives $|\bar{p}^{(A^*)} - W(A^* | x', s_0)| \geq \delta$ a.s. Picking $\xi = \delta/2$, the reverse triangle inequality in (??) yields

$$\Pr(|\hat{P}_{Y^n}(A^* | J_{\text{flip}}) - W(A^* | x', s_0)| < \delta/2) \leq 2e^{-N\delta^2/8}.$$

Since $\mathbb{V}(P, Q) = \sup_A |P(A) - Q(A)|$, taking $A = A^*$ on the LHS,

$$\Pr(\mathbb{V}(\hat{P}_{Y^n}(\cdot | J_{\text{flip}}), W(\cdot | x', s_0)) < \delta/2) \leq 2e^{-N\delta^2/8} \leq 2e^{-c_2 n},$$

where $c_2 \triangleq \alpha(1-\alpha)(1+\beta)\delta^2/8 > 0$ uses $N > \alpha(1-\alpha)(1+\beta)n$ from (iii). Combining (??) and (??), missed detection has probability $\leq 2e^{-c_2 n}$ for any single $\hat{m} \neq m$. Union-bounded over $N \leq 2^{nR}$ candidates, missed detection is $o(1)$ provided $R \ln 2 < c_2$.

Conclusion: Let c_1 and c_2 satisfy (??) and (??) respectively and let α, β satisfy (??). The above analysis shows that as long as R satisfies $R < \min\{c_1/\ln 2, c_2/\ln 2, \beta^2\alpha^2/6, \beta^2\alpha(1-\alpha)/4\}$, both error events are $o(1)$ uniformly over feedback-aware adversaries. $\square$

Lift to doubly-exponential rate via identification codes: We now lift the auth-code above to the doubly-exponential rate claimed in Lemma 15, mirroring the construction of [5]: compose the auth-code with an Ahlswede–Dueck identification code [25] for the noiseless binary channel.

Setup: Let W_{id} denote the noiseless binary channel ($\mathcal{X} = \mathcal{Y} = \{0, 1\}$, output equals input). Fix $R < R^{(\text{auth})}$ where $R^{(\text{auth})} > 0$ is the auth-code rate guaranteed by Appendix ?? . Pick $\hat{R} \in (R, R^{(\text{auth})})$ and let $\tilde{R} \triangleq R/\hat{R} \in (0, 1)$. Fix target errors $\lambda_1, \lambda_2 \in (0, 1/2)$.

Inner identification code: By [25, Theorem 1], for every $\tilde{n}$ large enough there is an $(N^{(\text{id})}, \tilde{n})$ -identification code $(\tilde{P}_{B^{\tilde{n}}|M}, \{\tilde{\phi}_m\}_{m \in [N^{(\text{id})}]})$ for W_{id} with misidentification errors $(\lambda_1/2, \lambda_2/2)$ and

$$N^{(\text{id})} \geq \lfloor 2^{2^{\tilde{n}\tilde{R}}} \rfloor.$$

Here $M \in [N^{(\text{id})}]$ is the source message, $\tilde{P}_{B^{\tilde{n}}|M}$ is the (stochastic) identification encoder, and $\tilde{\phi}_m : \{0, 1\}^{\tilde{n}} \rightarrow \{\text{ACCEPT}, \text{REJECT}\}$ is the candidate- m identification verifier. Its codeword set has size at most $N_{\text{out}}^{(\text{id})} \triangleq 2^{\tilde{n}}$.

Outer auth-code: By Appendix ??, for every n large enough there is an $(N^{(\text{auth})}, n)$ -authentication code $(\hat{P}_{X^n|B^{\tilde{n}}}, \hat{\Phi})$ for $W_{Y|X,S}$ with no-adversary error $\leq \min\{\lambda_1/2, \lambda_2/2\}$, missed-detection error $\leq \min\{\lambda_1/2, \lambda_2/2\}$ uniformly over feedback adversaries, and

$$N^{(\text{auth})} \geq \lfloor 2^{n\hat{R}} \rfloor.$$

The authentication code's message space is identified with the identification codeword space $\{0, 1\}^{\tilde{n}}$ by choosing n so that $N^{(\text{auth})} \geq N_{\text{out}}^{(\text{id})} = 2^{\tilde{n}}$ (achievable since $\hat{R} > 0$); the auth-code verifier $\hat{\Phi}(y^n) \in \{0, 1\}^{\tilde{n}} \cup \{\text{REJECT}\}$ either outputs a decoded identification codeword or rejects.

Composite tag: For each candidate $\hat{m} \in [N^{(\text{id})}]$, define an $(N^{(\text{id})}, n)$ -tag for $W_{Y|X,S}$ with encoder

$$P_{X^n|M}(x^n|m) = \sum_{b^{\tilde{n}} \in \{0,1\}^{\tilde{n}}} \hat{P}_{X^n|B^{\tilde{n}}}(x^n|b^{\tilde{n}}) \tilde{P}_{B^{\tilde{n}}|M}(b^{\tilde{n}}|m),$$

and verifiers

$$D_{\hat{m}}(y^n) = \begin{cases} \text{REJECT} & \text{if } \hat{\Phi}(y^n) = \text{REJECT}, \\ \text{REJECT} & \text{if } \hat{\Phi}(y^n) = \hat{b}^{\tilde{n}} \in \{0, 1\}^{\tilde{n}} \\ & \text{and } \tilde{\phi}_{\hat{m}}(\hat{b}^{\tilde{n}}) = \text{REJECT}, \\ \text{ACCEPT} & \text{otherwise.} \end{cases}$$

That is, the receiver first runs the auth-code verifier $\hat{\Phi}$ on Y^n to decode either an identification codeword $\hat{b}^{\tilde{n}}$ or REJECT; on a successful auth-decode it then runs the identification verifier $\tilde{\phi}_{\hat{m}}$ for the candidate $\hat{m}$ on $\hat{b}^{\tilde{n}}$.

Rate: By (??) and the requirement $N^{(\text{auth})} \geq 2^{\tilde{n}}$ (i.e., $\tilde{n} \leq (1/\hat{R}) \log N^{(\text{auth})} \leq n$),

$$\frac{1}{n} \log \log N^{(\text{id})} \geq \frac{\tilde{n} \tilde{R}}{n} \geq \frac{\tilde{n} \tilde{R}}{(1/\hat{R}) \log N^{(\text{auth})}} \geq \tilde{R} \hat{R} = R.$$

Thus $\frac{1}{n} \log \log N_n \geq R$, matching the rate claim of Lemma 15.

Error analysis: For the no-adversary side, $S^n = s_0^n$ implies $\hat{\Phi}(Y^n) = B^{\tilde{n}}$ except with probability $\leq \lambda_1/2$ (auth-code false alarm), and on that event the identification verifier $\tilde{\phi}_M(B^{\tilde{n}})$ accepts except with probability $\leq \lambda_1/2$ (identification false alarm). By union bound, the composite false alarm is $\leq \lambda_1$.

For the missed-detection side, fix $M = m$, $\hat{m} \neq m$, and any feedback-aware adversary Q . The composite verifier accepts $\hat{m}$ only if both (i) $\hat{\Phi}(Y^n) = \hat{b}^{\tilde{n}} \in \{0, 1\}^{\tilde{n}}$ (auth-code does not reject), and (ii) $\tilde{\phi}_{\hat{m}}(\hat{b}^{\tilde{n}}) = \text{ACCEPT}$ (identification verifier accepts). On (i), Lemma 15's auth-code half (proved in Appendix ??) guarantees that $\hat{b}^{\tilde{n}}$ equals the encoded $B^{\tilde{n}}$ except on an event of Q -uniform probability $\leq \lambda_2/2$. On the complementary event $\hat{b}^{\tilde{n}} = B^{\tilde{n}}$, the identification verifier $\tilde{\phi}_{\hat{m}}(B^{\tilde{n}})$ accepts with probability $\leq \lambda_2/2$ (identification missed detection). Union-bounding the two events,

$$\sup_Q \Pr(D_{\hat{m}}(Y^n) = \text{ACCEPT} \mid M = m) \leq \lambda_2 \quad \forall \hat{m} \neq m.$$

B. Composition with a channel code

We now provide full analysis for the composite encoder of Section V, addressing the subtlety that under the composite, the adversary in the tag phase may have *additionally* observed the wait-phase outputs Y^{n_1} .

Setup: Let $(F^{\text{ch}}, \phi^{\text{ch}})$ be a length- n_1 channel code over $W_{Y|X, S=s_0}$ at rate $R < C_{s_0}$ with no-adversary error $\epsilon_n = o(1)$. Let $(P^{\text{tag}}, \{D_{\hat{m}}^{\text{tag}}\})$ be an (N_n, n_2) -tag from Lemma 15 with errors (ϵ_n, ϵ_n) , $n_2 = O(\log n)$. The composite encoder is the product distribution (V-B); the composite decoder is $\phi(Y^{n_1+n_2}) = \hat{m}$ if $D_{\hat{m}}^{\text{tag}}(Y_{n_1+1}^{n_1+n_2}) = \text{ACCEPT}$ and REJECT otherwise, where $\hat{m} = \phi^{\text{ch}}(Y^{n_1})$.

No-adversary error: Under $S^{n_1+n_2} = s_0^{n_1+n_2}$, the codeword stage outputs the correct $\hat{m} = M$ with probability $\geq 1 - \epsilon_n$ and, on that event, the tag stage accepts with conditional probability $\geq 1 - \epsilon_n$ (since the tag's false alarm is $\leq \epsilon_n$). By a union bound, $\Pr_{\text{na}}(\phi(Y^{n_1+n_2}) \neq M) \leq 2\epsilon_n$.

Missed-detection error under feedback adversary: Fix any feedback-aware adversary $Q = \{Q_{S_t|S^{t-1}, Y^{t-1}}\}_{t=1}^{n_1+n_2}$. The composite missed-detection event is

$$\mathcal{M} \triangleq \{\phi(Y^{n_1+n_2}) \notin \{M, \text{REJECT}\}\},$$

i.e. $\hat{m} = \phi^{\text{ch}}(Y^{n_1}) \neq M$ and $D_{\hat{m}}^{\text{tag}}(Y_{n_1+1}^{n_1+n_2}) = \text{ACCEPT}$. We bound $\Pr_{\text{adv}}(\mathcal{M})$ via two steps.

Step 1 (decompose by $\hat{m}$). Conditioning on (M, Y^{n_1}) , the candidate $\hat{m} = \phi^{\text{ch}}(Y^{n_1})$ is determined; on the event $\hat{m} \neq M$, the tag-phase outputs $Y_{n_1+1}^{n_1+n_2}$ form a length- n_2 tag-channel transcript with encoder $P_{X^{n_2}|M}^{\text{tag}}$ and adversary states $S_{n_1+1}^{n_1+n_2}$ drawn according to $Q_{S_t|S^{t-1}, Y^{t-1}}$ restricted to the tag phase but with Y^{t-1} inheriting the wait-phase outputs Y^{n_1} as prefix.

Step 2 (induced adversary in tag phase). For each fixed wait-phase realization (M, Y^{n_1}, S^{n_1}) , define an *induced* tag-phase adversary

$$\tilde{Q}_{S_t|S_{n_1+1}^{t-1}, Y_{n_1+1}^{t-1}} \triangleq Q_{S_t|S^{t-1}, Y^{t-1}}(\cdot | (S^{n_1}, S_{n_1+1}^{t-1}), (Y^{n_1}, Y_{n_1+1}^{t-1})).$$

This $\tilde{Q}$ is a feedback-aware tag-phase adversary in the sense of Definition ?? – its t -th conditional uses only history through time $t-1$ within the tag phase, with the wait-phase realization frozen as a parameter. Lemma 15 states that for *every* such tag-phase adversary, $\Pr(D_{\hat{m}}^{\text{tag}} = \text{ACCEPT} \mid M = m, \tilde{Q}) \leq \epsilon_n$ for all $\hat{m} \neq m$. Marginalizing over (M, Y^{n_1}, S^{n_1}) under the composite attack,

$$\begin{aligned} \Pr_{\text{adv}}(\mathcal{M}) &= \mathbb{E}_{\text{adv}}[\mathbf{1}_{\hat{m} \neq M} \Pr(D_{\hat{m}}^{\text{tag}} = \text{ACCEPT} \mid M, \hat{m}, Y^{n_1}, S^{n_1})] \\ &\leq \mathbb{E}_{\text{adv}}[\mathbf{1}_{\hat{m} \neq M} \epsilon_n] \leq \epsilon_n. \end{aligned}$$

Total: The auth-error of the composite is the sum of the no-adversary error and the worst-case missed-detection:

$$\varepsilon(\phi) \leq \Pr_{\text{na}}(\phi \neq M) + \sup_Q \Pr_{\text{adv}}(\mathcal{M}) \leq 3\epsilon_n = o(1).$$

The composite rate is $\frac{nR}{n_1+n_2} = \frac{nR}{n_1+O(\log n)} \rightarrow R$, so $C_{\text{auth, stoch}}^{\text{fb}} \geq R$ for every $R < C_{s_0}$, finishing the achievability halves of Theorems 7 and 8. $\square$

APPENDIX E

PROOF OF DETERMINISTIC ACHIEVABILITY (THEOREM 9)

The deterministic achievability follows similar to the auth-code construction proved in Appendix ?? with P_X replaced by $\delta_{x_0}A$. We list the only places where the proof differs and verify that the rest carries over verbatim.

Witness: Use the I-overwritability witness (x_0, x'_0, δ) from (4). The distribution-overwritability gap (??) reduces to

$$\inf_{Q \in \mathcal{P}(\mathcal{S})} \mathbb{V}(\sum_s Q(s) W(\cdot | x_0, s), W(\cdot | x'_0, s_0)) \geq \delta,$$

which is exactly (4).

Encoder: Substituting $P_X = \delta_{x_0}$ in (??) makes the encoder fully deterministic: $X_t = x'_0$ for $t \in \mathcal{B}_m$, $X_t = x_0$ otherwise. The codeword $c_m \in \{x_0, x'_0\}^n$ is a deterministic function of m .

Step 1 of Appendix ?? (conditional law on the flip set): On J_{flip} the encoder transmits $X_t = x_0$ deterministically, so the conditional output law given $\mathcal{F}_{t-1}$ specializes to

$$\pi_t(y) = \sum_{s \in \mathcal{S}} Q_{S_t | \mathcal{F}_{t-1}}(s | \mathcal{F}_{t-1}) W(y | x_0, s).$$

Its flip-set average $\bar{P} = \sum_s \bar{Q}(s) W(\cdot | x_0, s)$ satisfies $\mathbb{V}(\bar{P}, W(\cdot | x'_0, s_0)) \geq \delta$ a.s. by the I-overwritability gap.

Step 2 (martingale concentration): Unchanged. The bounded-difference martingale $Z_t = \mathbf{1}_{Y_t \in A} - \pi_t(A)$ on J_{flip} (with π_t specialized as above) and the Azuma–Hoeffding bound (??) apply verbatim, since the $|Z| \leq 1$ envelope and the conditional-mean centering do not depend on the input distribution.

Step 3 (overlap property): Unchanged. The overlap-property bound (??) is a statement about codebook geometry alone and is identical for both cases.

Conclusion: The combining argument of Appendix ?? carries over verbatim, giving $\Pr_{\text{adv}}(\hat{M} \notin \{m, \text{REJECT}\}) \leq \lceil 2^{nR} \rceil \cdot 2|\mathcal{Y}|e^{-c_2 \mu n} = o(1)$ for $R \ln 2 < c_2 \mu$. Theorem 9's achievability follows. $\square$

APPENDIX F AZBSC VERIFICATION

The AZBSC(δ, α) has $\mathcal{X} = \mathcal{Y} = \{0, 1\}$, $\mathcal{S} = \{s_0, s_{1,0}, s_{1,1}\}$, and parameters $0 < \alpha < \delta < 1/2$. Under state s_0 , $W_{Y|X,S}(\cdot | \cdot, s_0)$ is a BSC(δ); under $s_{1,x}$, $W_{Y|X,S}(y | x, s_{1,x}) = \mathbf{1}\{y = x\}$; and

$$W_{Y|X,S}(y | x, s_{1,x \oplus 1}) = \alpha \mathbf{1}\{y = x\} + (1 - \alpha) \mathbf{1}\{y \neq x\}.$$

Distribution-overwritability. For $P_X(0) = p$ and target $x' = 0$, the strategy

$$Q(s_0) = 0, \quad Q(s_{1,0}) = \frac{1 - \delta - p\alpha}{1 - \alpha}, \quad Q(s_{1,1}) = \frac{\delta - (1 - p)\alpha}{1 - \alpha}$$

satisfies $\mathbb{E}_{X,S} W_{Y|X,S}(\cdot | X, S) = W_{Y|X,S}(\cdot | 0, s_0)$.

Failure of oblivious overwritability. An oblivious $Q(s_{1,0}) = q_0$, $Q(s_{1,1}) = q_1$ matching $W_{Y|X,S}(\cdot | 0, s_0)$ for both $x = 0$ and $x = 1$ forces $q_0 + q_1 = (1 - 2\delta)/(1 - 2\delta - \alpha) \notin [0, 1]$ for any admissible (δ, α) .